

# Introduction to Probability and Statistics

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# Topics Covered

## Introductions

- Name, degree sought, full-time/part-time, work experience
- Research and teaching activities

## Syllabus

- Office hours, readings, grading, etc.

## Topics

- Statistics in the real-world
- Course overview
- Types of data and levels of measurement
- R/RStudio
- Artificial intelligence

# Syllabus in a Nutshell I

## Office hours

- By appointment via email
- Come to office hours early in the semester
- Step learning curve regarding R/RStudio

## Readings

- No required textbook but potential reference books available for free through the library
- Not related to R but to probability and statistics more generally: [A Modern Introduction to Probability and Statistics](#)

# Syllabus in a Nutshell II

## Assignments and grading

- 7 assignments to be submitted each as one PDF
- Grading on linear scale without curve to avoid grading relative to classmates

## Online course evaluations

- You receive 2 percentage points on your final grade if more than 90% of students fill out the online course evaluation at the end of the semester.

No restriction on the use of artificial intelligence (more on that later)

# Risk and Uncertainty in Everyday Life

## Grades

- Uncertainty surrounding class grade during a semester
- Association of probabilities with each grade

## Fire station calls

- Number and location of calls
- Number of fire trucks and other vehicles required

Two outcomes does not mean a 50% chance for each to happen

- Success of a free throw by Stephen Curry
- Flight delay due to fog

# Statistics in the News

## Election outcomes

- 2002 French Presidential Election
  - Two-stage election
  - Final round: Jacques Chirac (82.2%) and Jean-Marie Le Pen (17.8%)
- 2016 U.S. Presidential Election
  - [FiveThirtyEight](#) forecast of Donald Trump winning: 28.6%
  - [Cognitive biases versus data as an explanation](#)
- 2024 U.S. Presidential Election
  - 7 swing states that all voted for Donald Trump: Chance of 0.78% of that outcome in the case of independence of events

## [Path of hurricane Sandy](#)

# Financial Economics

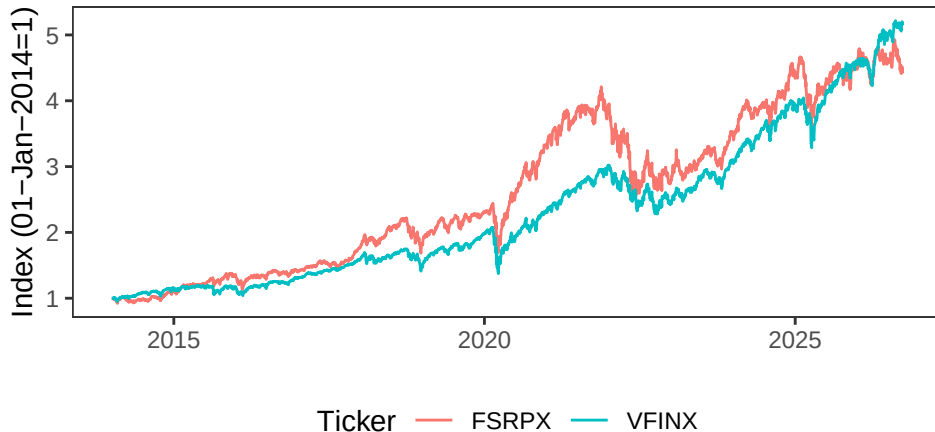
## Evolution of the stock market

- Importance of correlation among stocks and mutual funds

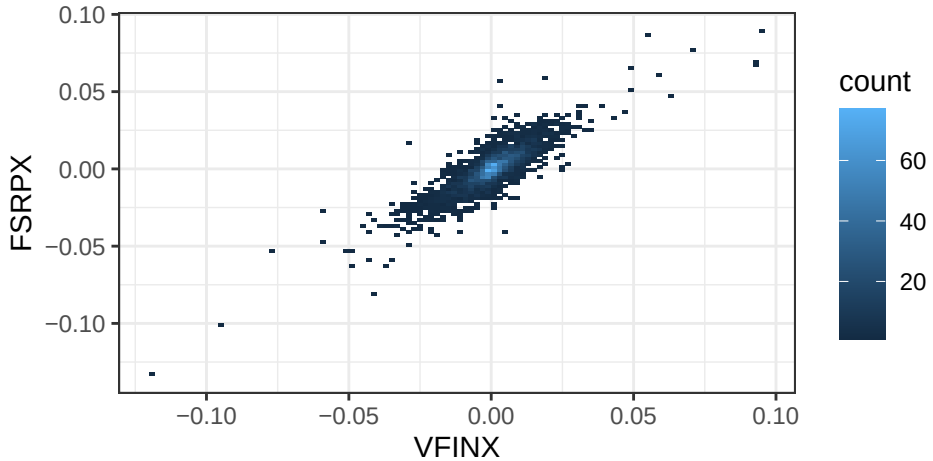
## Next slides

- FSRPX: Fidelity Select Retailing Portfolio
- VFINX: Vanguard 500 Index Fund Investor Shares

# FSRPX and VFINX: Evolution



# FSRPX and VFINX: Returns Scatter Plot



# Scottish Ministers' Widows' Fund

## Preceding work

- Edmond Halley's (same as comet) life tables for the city of Breslau (today Wrocław) in 1693
- Detailed work on birth and death by age

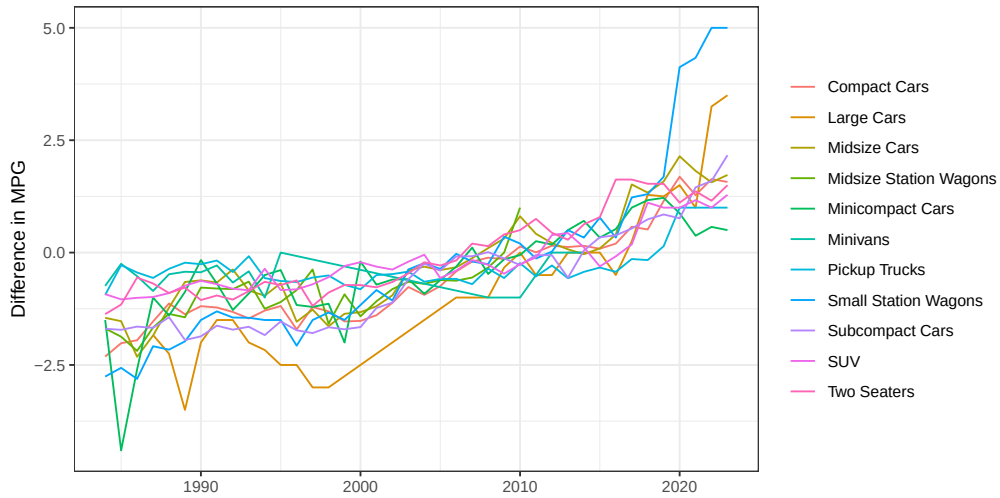
## Insurance fund calculations in 1744 by Alexander Webster and Robert Wallace

- Payments to widows and heirs after death of ministers
- Required information: Number of clergymen, deaths per year, life expectancy of surviving family, time of remarriage, etc.
- Calculation of annual payments into the fund

## Fund balance (in pound sterling) in 1765

- Estimated: 58,348
- Actual: 58,347

# Automatic vs. Manual Transmission



# Data, Probability, Statistics, and Regression

## Data and data visualization

- Descriptive statistics and graphical presentation of data

## Probability

- Providing basis for modeling populations, experiments, and any other random phenomena

## Statistics

- Learning about the population based on a sample

## Regression analysis

- Mathematical relationship among variables

Difference between probability and statistics: Bucket example

# Data and Descriptive Statistics Overview

## Data

- Raw observations collected from surveys, experiments, or administrative records.
- Examples in public policy: Annual household income, homicide rates, citizens' opinions on a policy

## Descriptive statistics

- Summarizes and simplifies data to reveal patterns
- Examples: Average donations per year to a nonprofit
- Common descriptive statistics are related to central tendency (e.g., mean, median) and dispersion, i.e., spread of the data measured by range, variance, or standard deviation
- Purpose: Understand the typical values, detect variability and outliers, and provide a foundation for further analysis

# Data Visualization Overview

Graphical representation of data to make patterns intuitive and interpretable

- Increasing importance for communicating with the public
- Examples previously used: Pattern of financial returns and fuel economy

Common techniques used

- Histograms, bar charts, boxplots
- Scatterplots, line charts

Benefits:

- Identify trends, relationships, and anomalies
- Communicate results clearly to policymakers and stakeholders

# Probability Overview

## Study of uncertainty and randomness

- Given a model, what are the chances of an event happening

## Public policy examples

- Chance (probability) of a voter turnout of over 60%
- Probability of extreme weather events

## Topic covered

- Probability theory: Many examples of flipping coins and rolling dice
- Probability distributions: How to model random outcomes

## Foundation for statistics

# Difference between Population and Sample

## Population

- A population is the collection of all possible individuals, entities, objects, or measurements of interest for a particular investigation. A sample is any portion or subset of the population. A *parameter* characterizes the population and is usually unknown (forever).

## Sample

- A statistic is any measurable characteristic of a sample. Statistical analysis utilizes statistics from representative samples to infer the parameters of an entire population.

## Using a sample rather than the population

- Cost considerations
- Possible destruction of observation units (e.g., mileage of tires)
- Unfeasible to study all units of observations

# Statistics Overview

## Inference

- Given data, what can we infer about the population or phenomenon?

## Examples:

- Estimating average income in a city
- Calculating unemployment rates from survey data
- Relies on probability to quantify uncertainty in inferences

## Topics covered

- Sampling
- Confidence intervals
- Hypothesis testing

# Regression Overview

## Regression analysis

- Statistical method to model the relationship between a dependent variable ( $Y$ ) and one or more independent variables ( $X$ )
- Correlation is not causation!

## Types of regression models

- Simple linear regression (i.e., one independent variable), e.g., effect of education on income
- Multiple regression (i.e., multiple independent variables), e.g., effect of education, experience, and age on income
- Other regression models: Logistic regression (binary outcomes), Poisson regression (count data), etc.

Example: Price of a used car as a function of mileage

# Variables

## Qualitative variables

- Non-numeric, e.g., gender, political affiliation, state of residence
- Can be transformed into numerical value, i.e., “dummy variables” in regression analysis

## Quantitative variables

- Numeric, e.g., age, income, GPA, number of kids

## Quantitative variables can be either

- Discrete: Take two close values and there is no value in between, e.g., number of people in a class
- Continuous: Take two close values and there is always (!) a value in between, e.g., weight of a people

# Levels of Variable Measurements

## Nominal

- Categories, e.g., eye color, gender, religious affiliation, mode of transportation to O'Neill IU Indianapolis
- No natural ordering

## Ordinal

- Categories, e.g., level of happiness, Homeland Security Advisory System
- Natural ordering, i.e., data can be ordered

## Interval

- Intervals between levels are equally spaces and differences between variables have a meaning
- Examples: Income, GPA, etc.
- Most commonly used in this class.

# Introduction to R and RStudio

## R

- A programming language and environment for statistical computing and graphics
- Widely used in public policy, economics, and data science
- Open-source and free

## RStudio

- An integrated development environment (IDE) for R
- Provides a user-friendly interface with: Script editor, console, environment/history panels as well as plots, files, and packages panels

## Reasons to use R and RStudio

- Powerful statistical and graphical capabilities
- Active community with thousands of packages for specialized analyses
- Reproducible research with R Markdown and Shiny

# Advantages and Disadvantages

## Advantages

- Free and open-source
- Supports advanced statistics and machine learning
- Excellent for data visualization and reporting
- Reproducible workflows with R Markdown (do not use for homework)
- Active and helpful community
- Maybe most importantly: Easy to use with AI

## Disadvantages

- Steeper learning curve than spreadsheet software
- Some packages may have inconsistent syntax
- Memory-intensive with very large data sets
- Limited GUI support compared to commercial software (e.g., SPSS, Stata)

# Artificial Intelligence

## AI is built on statistics

- AI models are essentially statistical models trained on large data sets
- They estimate patterns, probabilities, and relationships from data

## Learning from data

- AI learns from samples to make predictions about unseen cases.
- Example: ChatGPT predicts the next word based on previous words using conditional probabilities.

## Probabilities everywhere

- Every AI prediction involves calculating probabilities of possible outcomes.
- Uncertainty is inherent: multiple valid responses can exist for the same input.